<u>Practice Problems List-14 (Structures)</u>

1. Define a structure named **Person** with members *name* and *age*. Input these values and print them.

2. Define a structure named **Person** with members *name* and *age*. Swap the values of two variables of **Person** type.

3. Define a structure **Date** with members day, month, year. Store the birthdays of two people and output which one is older.

4. Define a structure **Point** with members x and y. Initialize a Point structure directly and display which quadrant it falls into.

5. Define a structure **Point** with x and y. Check if two Point structures are equal by comparing their members. Implement a function **`isEqual()`** to solve the problem.

6. Create a structure named Book to store book details like title, author, genre, and price. Write a C program to input details for five books, find the most expensive and the lowest priced books, and display their information. Create two separate functions **`displayCostliest`**() and **`displayCheapest`**() to solve the problem.

7. Create a structure named "Employee" to store employee details such as employee ID, name, and salary. Write a program to input data for three employees, find the highest salary employee, and display their information.

8. Create a structure called "Student" with members name, age, and total marks. Write a C program to input data for 45 students, display their information, and find the average of total marks.

9. Create a structure named "Employee" to store employee details such as employee ID, name, birth date, and salary. The birth date should be of another structure **Date** with members day, month, year. Write a program to input data for five employees. Implement a function **`displayEmployeeDetails`**() to solve the problem.

10. Define a structure Sample with members int x, char y, and float z. Print the size of the structure.

11. Define two structures named **Triangle** with members a, b, and c, signifying the three sides of the triangle. Define another structure named **Measurements** with members area and circumference. Assign values to a variable of **Triangle** type, calculate the area and circumference, and assign those measurements to a variable of **Measurements** type.

12. Solve the previous problem using a function named **`storeMeasurements`**().

13. Define a structure named **Person** with members *name* and *age*. Input the values using pointers, and print them.

14. Solve the previous problem using a function named **`printPerson`**() that accepts the address of **Person** type.

15. Define a structure named **Person** with members *name* and *age*. Swap the values of two variables of **Person** type using pointers.

16. Solve the previous problem using a function named **`swapPerson`**() that accepts the addresses of two Person variables.